Thermodynamic principles beyond physical systems-I

(Dated: 9 November 2026)

Statistical mechanics is a very remarkable formulation in physics that allows one to understand the behavior of a collection of objects and how energy flows across these objects. However, standard texts and popular parlance undermine a true beauty hidden in the mathematical structure of statistical mechanics. We note that mathematics is dimensionless and if it is used to describe physics, it is likely that there are other scenarios where those concepts could apply. This philosophy inspired this work where we first develop an abstract notion of various thermodynamical ideas and then use it on a system of dice, which is usually only seen as an uninteresting system with uniform distribution. In this system, we demonstrate the emergence of Boltzmann distribution, the existence of a 'temperature', the notion 'thermal equilibrium', the possibility of 'heat transfer', and a natural emergence of second law of thermodynamics. This is part-I of a two-part article where we focus on the first three demonstrations.

CONTENTS

I. Introduction 2

II. The dice rolls 4

III. The Boltzmann distribution 5

A. Some definitions 5

B. Entropy and its maximization 6

C. A case of identical components 8

D. Connection to the terminology of conventional statistical physics 9

IV. Understanding the dice result 10

A. Why does the dice act like a system with maximized entropy? 12

V. Conclusion 14

A. Mapping the statistical ideas to information theory elements 15

References 17

I. INTRODUCTION

Laws of thermodynamics, as introduced in textbooks, are presented as the governing principles that dictate how a collection of many bodies behave. These laws have been formulated and established by careful experimentation and meticulous analysis (see Ref.[1] for a historical overview). Accepting these laws helps us understand and engineer energy flow around us, and discover new material properties by understanding phase transitions. One cannot emphasize enough the true beauty of these ideas which lies in the fact that it lays down the unavoidable rules that govern the energy flows irrespective of the details of $\sim 10^{23}$ constituents of system. In fact, the extreme utility despite the ignorance of details is the true charm of these ideas.

Although thermodynamics was established by the early 1800s (through foundational works of Carnot[2], Joule[3], and others), and formalized in the mid 1800s by Clausis[4] and Kelvin[5], a purely statistical formulation was developed, consolidated, and proliferated by the works of Maxwell, Boltzmann, and Gibbs by late 1800s[6]. A central theme to these latter ideas was that everything

was some sort of a counting problem. Nevertheless, it was still the physical systems that were the main target on which these ideas were tested. The next push was pioneered by Jaynes who maintained that most if not all concepts can be derived from the maximum entropy principle if one can correctly formulate the constraints on your system[7,8]. Indeed, the Boltzmann distribution, the Gaussian distribution (and a version of the central limit theorem), the Pareto distribution, can all be effectively deduced from the maximum entropy principle[9].

This article, is presented largely in the spirit of Jaynes' philosophy in that it explores a pure mathematical, and hence abstract, formulation of the thermodynamical ideas of energy, temperature, micro- and macro-states, equilibrium, heat transfer and the approach towards equilibrium. The abstraction helps us understand how the maximum entropy formulation comes to be and thus help extend these ideas to other systems. It is, nevertheless, distinct from Jaynes' approach in that it takes a frequentist's approach rather than the epistemological one advocated for by Jayne's. But our hope is that by the end of these two articles we can begin to see a connection between the two.

We present our perspective in two parts. This article, the first part, is an elaboration of the Brandeis dice problem popularized by Jaynes[10,11]. In Jaynes' formulation this problem appeared in the context of drawing an inference based on a given observation, but here we present it in the context of a system trying to attain or being in 'equilibrium' (we explain this term precisely in the text), thereby developing some harmony between the two perspectives. This part is organized in the following manner. Section II introduces a statistical system of dice that, surprisingly demonstrates Boltzmann's statistics. Section III presents a natural, yet purely statistical, perspective of popular statistical physics ideas of microstates, macrostates, entropy, and equilibrium, without explicitly referring to them. It arrives at the Boltzmann distribution and the notion of maximizing entropy on purely statistical grounds. In Sec. IV we map the abstract perspectives to the dice problem and explain why the system of dice acts like a system in equilibrium with a thermal bath. We will end this part by extending this perspective to the notion of entropy in information theory.

In part two of the article, we will push the analogy further. As stated above, thermodynamics also dictates how energy flows between systems to attain equilibrium. But if thermodynamics is, after all a statistical concept, it should be applicable to any statistical system, like our system of dice. Here we first work through the notion of attaining equilibrium on purely statistical terms and then demonstrate how a system of dice could be used to illustrate that idea. The real motivation behind this set of articles is drawn from historical and recent discussions[12–14] that posit that physics should essentially be dimensionless and hence the ideas contained in physical laws can,

in controlled circumstances, be applied to other systems (such as machine learning, information theory, biological systems, economic theories, etc.). In this set of works, we chose to demonstrate it on a set of dice which are successively rolled.

II. THE DICE ROLLS

Consider a collection of $D = 10$ fair dice (with faces $\{1, ..., 6\}$). We throw them $N_T = 100,000$ number of times and collect the sum of faces Σ at each turn and plot a histogram of the sum. The result is shown in Fig. 1(left), where the $\Sigma_{\text{min}} = 10$ and $\Sigma_{\text{max}} = 60$. This histogram is entirely predictable from the knowledge of combinatorics. The number of ways you can get a sum Σ is given by the coefficient of x^{Σ} in

$$(x^1 + x^2 + x^3 + x^4 + x^5 + x^6)^{DN_T}.$$

But we can ask a different question: for a given Σ, what is the distribution of the number of times a given face shows up. Estimating this computationally is simple. We just sample the subset of the N_T throws that gave Σ and perform a count of the faces. The result is shown in Fig. 1(right) for $\Sigma = 20, 30, 40, 50$. At the first glance, the histogram's generic feature that for $\Sigma < \Sigma_{\text{avg}} = 35$, the lower faces should occur more frequently (and vice versa), is entirely comprehensible. What is less so is the fact that these histograms are exponential, and more precisely they are Boltzmann like ($\sim e^{-\beta x}$). Figure 1(right) also shows the fit of the exponential curve to the histograms for specified Σ's.

The obvious question is why does a system of dice have a Boltzmann distribution which is so often attributed to physical systems with physical energies. An experienced reader will immediately answer that this is because specifying Σ plays the role of energy constraint on the system, where energy here can be identified with the (sum of the) face values of the dice. This is the solution to the Jaynes' Brandeis dice problem[11]. This answer is certainly correct, but the Boltzmann distribution is a solution to a system in equilibrium, with a bath. The above association with Jaynes' solution does not explain what is the notion of equilibrium for the dice, nor does it specify the bath it is in equilibrium with. Without a concrete correspondence of these additional features, the Boltzmann distribution appears to be coincidental.

This article is aimed at convincing both experienced and novice readers that this is not a coincidence. But first, it will be necessary to revisit the basics of statistical physics. There are many

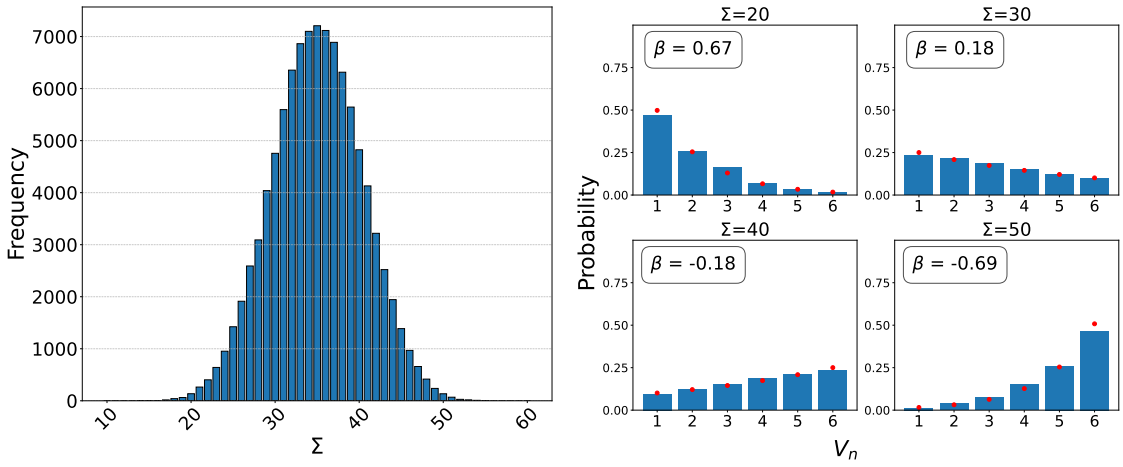


FIG. 1. Left: Histogram of all the possible sums Σ observed over $N_T = 100,000$ throws of $D = 10$ fair dice. Right: Probability (relative frequency) of face values $V_n \in \{1,2,3,4,5,6\}$ for the subset of throws where $\Sigma = 20,30,40,50$. The red dots denote $p_n = e^{-\beta V_n}/Z$, where $Z = \sum_n e^{-\beta V_n}$. The respective values of the parameter β is presented in the plot.

textbooks that we can refer the reader to, but we choose to present our version wherein we will try to emphasize aspects that are generalizable beyond physical systems with conventional energy. In fact, we will avoid the textbook terminology (energy, micro- and macro- states) so as to discourage the readers from exercising their biases in reading into something that is not stated.

III. THE BOLTZMANN DISTRIBUTION

We start by introducing our terminology. When necessary, we will connect them to the conventional ones.

A. Some definitions

Components and configurations of our system: Consider a system made out of certain components. Each component can be in a certain state s_n, where n spans the dimension of the state space which we label with $\mathscr{P}_S$. The union of the states of these components makes up a configuration of the system. The phase space of all configurations will be called the configuration-space $\mathscr{P}_C$. The evolution of the system (according to whatever rule) is then simply a traversal through this space. Then, for an observation period τ_{Obs}, there emerges a certain probability p_i for the system to be in

a configuration $c_i \in \mathscr{P}_C$[15], such that $\sum_i^{N_{\mathscr{P}_C}} p_i = 1$, where $N_{\mathscr{P}_C}$ is the number of configurations in the space $\mathscr{P}_C$. The probability itself can be estimated as $p_i = \tau_i/\tau_{\text{Obs}}$, where τ_i is the time (does not have to be contiguous) spent in configuration c_i. In fact, the notion of time can be replaced by a sequence or even a collection of observations where τ_{Obs} would map to the total number of observations.

Value-based states of our system and their observations: Let us now associate a value V_i to each configuration c_i[16]. Also consider an observation protocol which over a period τ_{Obs} can only observe the cumulative (average) value associated with the system. That is, over τ_{Obs}, the protocol detects a number V, which can be related to the component/configuration variables as $V = \sum_i^{N_{\mathscr{P}_C}} p_i V_i$. For a given value prescription $\{V_i\}$, the value V is linked to the particular evolution of the system that is characterized by $\{p_i\}$. If the observed V remains stationary irrespective of τ_{Obs}, we will say that the system is in a *derived state* with value V (as it is derived from the detailed configuration-state of the system).

B. Entropy and its maximization

If we only have access to the value V of the system observed over τ_{Obs}, we may be interested in asking what evolution would lead to the $\{p_i\}$ that produces the value V. There usually are many different $\{p_i\}$'s that could result in the same observation of V. With no other knowledge of the component-level information of the system, the best guess would be the distribution $\{p_i\}$ that is most likely (where the system spends most amount of time in).

To go about finding this most likely distribution, consider taking $\mathscr{N} \gg 1$ snapshots (observations) of the configurations of the system over τ_{Obs}. We can then count the number of snapshots that correspond to configuration c_i of the system: $\mathscr{N}_i$, such that $\sum_i \mathscr{N}_i = \mathscr{N}$. With $\{p_i\} \equiv \{\mathscr{N}_i/\mathscr{N}\}$, there are $\mathscr{W} = \frac{\mathscr{N}!}{\mathscr{N}_1!\mathscr{N}_2!\ldots\mathscr{N}_{N_{\mathscr{P}_C}}!}$ possible *instances of observations* that would support the specified $\{p_i\}$.

To understand how different instances can arise, consider a long observation time $\tau_L = \mathscr{W}_0 \tau_{\text{Obs}}$, where $\mathscr{W}_0 \gg \mathscr{W}$. In each unit of time τ_{Obs}, one takes the $\mathscr{N}$ snapshots. This long observation time offers the opportunity to realize all possible evolutions that one can observe during various chunks of τ_{Obs}. It is then sensible to claim that the most likely $\{p_i\}$ to be inferred from each chunk of $\mathscr{N}$ snapshots will correspond to the one with the highest multiplicity in the $\mathscr{W}_0 \tau_{\text{Obs}}$ observation time. Thus, we should seek $\{\mathscr{N}_i\}$ that maximizes $\mathscr{W}$. Given no other constraint, deploying combinatoric

principles, we know that $\mathscr{W}$ is maximized when $\mathscr{N}_i = \mathscr{N}/N_{\mathscr{P}_C}$. That is, all configurations are equally accessed. In the limit of large $\mathscr{N}$, instead of working combinatorially, one can introduce a mathematical formulation to find Max$[\mathscr{W}]$. Since the function ln is monotonic, finding Max$[\mathscr{W}]$ is the same as finding Max$[\ln\mathscr{W}]$:

$$\ln\mathscr{W} \xrightarrow{\mathscr{N}\gg 1} -\mathscr{N}\sum_i \tilde{p}_i \ln\tilde{p}_i \equiv \mathscr{N}\tilde{S}[\{\tilde{p}_i\}], \tag{1}$$

where $\tilde{p}_i = \mathscr{N}_i/\mathscr{N} \geq 0$. The maximization is done by introducing a Lagrange multiplier to implement the constraint $\sum_i \tilde{p}_i = 1$. The result that maximizes $\ln\mathscr{W}$ is $\tilde{p}_i = \mathscr{N}_i/\mathscr{N} = 1/N_{\mathscr{P}_C}$. This is the same as the combinatorial result. The functional $\tilde{S} = -\sum_i \tilde{p}_i \ln\tilde{p}_i$ is popularly called the entropy, and maximizing it yields a distribution that is maximally likely to be observed when no other limitations are placed. Observe that the number of snapshots $\mathscr{N}$ is just a scaling factor for the (log of the) multiplicity. This is actually a curiously noteworthy result of an important property of the entropy functional defined in Eq. (1): the functional is additive for independent distributions. That is, for an independent $\{p_i\}$ and $\{q_j\}$ (even with different phase space sizes) the definition ensures

$$\tilde{S}(\{p_i\},\{q_j\}) = \tilde{S}(\{p_i\}) + \tilde{S}(\{q_j\}). \tag{2}$$

One could argue that the choice of maximizing $\ln\mathscr{W}$ instead of $\mathscr{W}$ itself was made to exploit this very desirable additive property[17]. This property ensures that for identical, independent distributions (iid's) the total entropy is simply a multiple of the entropy of the base distribution. The notion of a base distribution essentially posits that the likelihood of even a single observation also reflects the likelihood over multiple observations (made with identical protocol). This offers the notion of a time-average (sequence of observations) being equal to the phase-space (ensemble) average for a single observation. Returning to obtaining the largest multiplicity of a particular distribution, which is just the scaled version of the maximum entropy, we find that

$$\tilde{S}_{\max} = \ln N_{\mathscr{P}_C}, \tag{3}$$

where $N_{\mathscr{P}_C}$ is the size of the phase-space. The implication of Eq. (3) will be discussed in Sec. III D.

Let us now, from the above evolution (snapshots), only retain those observations that correspond to a particular value V. The most likely $\{p_i\}$, subject to this value-based constraint, will be different from the unconstrained one. This can be found by solving for

$$\text{Max}[\tilde{S}(\{\tilde{p}_i\})] \text{ such that } \sum_i \tilde{p}_i = 1,\ \sum_i \tilde{p}_i V_i = V. \tag{4}$$

This is again achieved by introducing a Lagrange multiplier for each constraint. Labeling the multiplier for the value-constraint as β_V, the result can be written as

$$\tilde{p}_i = \frac{1}{Z} e^{-\beta_V V_i}, \text{ where } Z \equiv \sum_i e^{-\beta_V V_i}, \tag{5}$$

and β_V is found from the transcendental equation

$$\frac{1}{Z} \sum_i V_i e^{-\beta_V V_i} = V. \tag{6}$$

If $\beta_V = 0$, then the multiplier for the constraint vanishes freeing the system from the value-based constraint and reproducing the unconstrained result of a uniform distribution.

For a given value of V, the maximized entropy $\tilde{S}_{\max} \equiv S$ is

$$\begin{aligned} S &= \beta_V V + \ln Z \\ &= \ln Z e^{\beta_V V} \\ &= \ln \bar{Z}, \end{aligned} \tag{7}$$

where $\bar{Z} \equiv \sum_i e^{-\beta_V (V_i - V)}$ is similar to Z but the V_i's are measured relative to V. Note that $\bar{Z} = N_{\mathscr{P}_C}$ if either $\beta_V = 0$ (no constraint) or $V_i = V$ (all values are the same, which is effectively disregarding the constraint). Then $S = \ln \bar{Z} = \ln N_{\mathscr{P}_C}$, reproducing the result of Eq. (3). Thus, the quantity $\bar{Z}$ can be seen as the size of the phase-space consistent with the constraints on the system with $e^{-\beta_V (V_i - V)}$ serving as reduction (if $V_i > V$) or enhancement (if $V_i < V$) of the relative frequency of sampling of those configurations relative to the unconstrained case. Because $\bar{Z}$ re-partitions the sampling frequency of the space $\mathscr{P}_C$, it is also called the *partition function* for the observation of V. Like in Eq. (1), the maximized value of $\tilde{S}$ still bears the interpretation of the log of the multiplicity of the distribution, but this time being consistent with the constraints.

C. A case of identical components

If the system components are identical, then they all share the same state space $\mathscr{P}_S$. In such a case, the space $\mathscr{P}_C$ is simply $(P_S \otimes)^D$, where D is the number of components. Additionally, if the value prescription is associated directly with s_n and not with any interdependency on the states of individual components, we can say that the components are identical, independent, and the value prescription is non-interfering. In such a case, we can write $V_i = \sum_n V_n N_n^i$, where N_n^i is the number of components in state s_n in the configuration c_i. This can be rewritten as $V_i =$

$D\sum_n V_n N_n^i/D = D\sum_n V_n p_n^i$, where $p_n^i \equiv N_n^i/D$ is the probability of finding a component in state s_n for the configuration c_i. In this form, V_i for the configuration c_i can be expressed in terms of the expected value of a single component times the number of components.

Now, when we filter out the observations to only retain V, all of V_i's are just V and p_n^i become independent of i. Then we have,

$$\begin{aligned} V &= \sum_i p_i V_i \\ &= \sum_i p_i (D \sum_n V_n p_n^i) \\ &= \sum_i p_i (D \sum_n V_n p_n) \\ &= D \sum_n V_n p_n. \end{aligned} \tag{8}$$

That is, a constraint on the value of the full system via V, in this case of independent identical components, is the same as a constraint on a single component via the value V/D. The entropy functional can be expressed as $\tilde{S}(\{p_i\}) = D\tilde{S}(\{p_n\})$, which simply is D times the entropy of a single dice. This simply reflects the additive property of entropy highlighted in Eq. (2). Thus, instead of using $\{p_i\}$ (the distribution of configurations) to define entropy, one can use $\{p_n\}$ (the distribution of states). Thus, the entropy maximization problem is

$$\text{Max}[\tilde{S}(\{\tilde{p}_n\})] \text{ such that } \sum_n \tilde{p}_n = 1,\ \sum_n \tilde{p}_n V_n = \frac{V}{D}, \tag{9}$$

where $\tilde{S}(\{\tilde{p}_n\}) = -\sum_n \tilde{p}_n \ln \tilde{p}_n$. The summations are now over the much smaller space $\mathscr{P}_S$. This reduction in the phase space if often extremely useful to estimate the multiplicity: $\ln \mathscr{W} \approx \mathscr{N}\tilde{S}(\{\tilde{p}_i\}) = \mathscr{N}D\tilde{S}(\{\tilde{p}_n\})$.

D. Connection to the terminology of conventional statistical physics

The experienced reader must have already identified that various configurations of the system in terms of its components are the microstates of the system as introduced in statistical physics. The value-based derived states are the macrostates, with the value being energy. The parameter β_V, that we have not explored yet, plays the role of (inverse) temperature. A less apparent connection is the concept of the sampling that filters out observations not consistent with the constraint. In physical systems, the conservation of energy serves as this filter. Also in the conventional scenario, it is never realistically possible to track the micro-states of the system. Most observations are only

at the level of the macro-state characterized by the energy of the system. The micro-states of the system are only *inferred* via a validation of the value-prescription through observations of the derived states (macro-states) of the system.

Something that is not often emphasized when the concept of entropy is introduced is the additive property of the functional. This is, in fact, what allows one to define entropy in a rather flexible manner, especially in the presence of independent an identical constituents/observations, where the phase space can either be the immensely vast configuration space of a many-body system, or just the simpler individual states of its constituents. Such definitions, taken individually, could give the impression of different definitions of entropy in different phase spaces, but we can hopefully see that they are systematically related. What becomes crucial in selecting a definition is the associated value prescription.

Next, returning to Eq. (3) we see that $S = \ln N_{\mathscr{P}_C}$. Having mapped the configuration space to micro-states of the system, $N_{\mathscr{P}_C}$ is the number of micro-states of the system. One recognizes this as the Boltzmann's definition of entropy (up to the factor of the Boltzmann's constant). In the case of identical components, this becomes $S = D \ln N_{\mathscr{P}_S}$. However, sometimes the explicit discussion of the additive property is omitted, resulting in the identification: $S = \ln N_{\mathscr{P}_S}$, which at its face is a different definition from $S = \ln N_{\mathscr{P}_C}$, but we can see the connection between the different definitions. However, the definition Eq. (1) and the Boltzmann's version $S = \ln N_{\mathscr{P}_C}$ are not the same definitions. The former applies to any distribution while the latter is a result for equilibrium state respecting the constraints.

Finally, we note that the state with maximum entropy has the maximum multiplicity of the microstates. This means that a microstate-level observation of the system will most likely find the system in this entropy maximized set of states. This is particularly true for very large systems where for they can be 'locked-in' in the most likely set of configurations characterized by $\{p_i\}$ (or $\{p_n\}$) for a very long time (large number of observations). This locking-in maps to the notion of equilibrium in physical systems. That is, in the abstract sense that is relevant here, we see that this notion of equilibrium is simply the maximized entropy state respecting all constraints.

IV. UNDERSTANDING THE DICE RESULT

With the abstract understanding of Sec. III, we are in a position to formally map the dice parameters to the various elements of the entropy maximization principle. The dice are the components

($D = 10$) that make up our system. Each die has access to s_n ($n \in \{1,...,6\}$) possible states which belong to $\mathscr{P}_S = \{1,2,3,4,5,6\}$. The configuration of the system is the set $c_i = \{N_1^i,...,N_6^i\}$, such that $\sum_n N_n^i = D$ and $N_n^i \geq 0$. This is just the collection of number of dice in each state in $\mathscr{P}_S$. The total number of configurations $N_{\mathscr{P}_C}$ is 6^D. We note that all dice are equivalent, which means, the entropy maximization could be applied at the level of the $\mathscr{P}_S$ space instead of $\mathscr{P}_C$. The value prescription is to sum the face value $V_n \in \{1,2,3,4,5,6\}$ of each die. The specification of Σ amounts to specifying the requirement of expected value of a single dice to be Σ/D. Finally, the N_T throws maps to the $\mathscr{N}$ snapshots of the system.

With this mapping, it is immediately clear from discussions in Sec. III that if no constraints are placed on the expected value, then $\tilde{p}_n = 1/N_{\mathscr{P}_S} = 1/6$. This is seen in Fig. 2 where we take all the configurations in the histogram in Fig. 1 and plot the frequency of occurrence of the faces (number of times V_n shows up over the DN_T instances). If one specifies a Σ, then

$$\tilde{p}_n = \frac{1}{Z} e^{-\beta V_n},\ Z = \sum_n e^{-\beta V_n}, \tag{10}$$

where β is found form solving $\sum_n p_n V_n = \Sigma/D$. This is the distribution shown in red dots in Fig. 1(right).

A hidden symmetry: In Fig. 1(right) observe that for $\Sigma = 40$ and $\Sigma = 50$ the distribution is a mirror of the $\Sigma = 30$ and $\Sigma = 20$, respectively, with the corresponding β's related by a minus sign. This is, in fact, the manifestation of a symmetry in Boltzmann's distribution that is less often discussed. Let us rewrite Eqs. (5) and (6) as

$$\begin{aligned} p_i \ &= \ \frac{1}{\bar{Z}} e^{-\beta_V (V - V_i)},\ \text{where } \bar{Z} = \sum_i e^{-\beta_V (V_i - V)}, \\ &\beta_V \text{ is got from } \frac{1}{\bar{Z}} \sum_i (V_i - V) e^{-\beta_V (V_i - V)} = 0. \end{aligned} \tag{11}$$

Note the invariance with respect to the following changes:

- Inversion: $\beta_V \to -\beta_V$ and $\{\{V_i\}, V\} \to \{\{-V_i\}, -V\}$.
- Change of reference: $\beta_V \to \beta_V$ and $\{\{V_i\}, V\} \to \{\{V_i + \delta V\}, V + \delta V\}$.

The negative β_V results in Fig. 1(right) is a consequence of $\beta_V \to -\beta_V$ and $\{\{V_i\}, V\} \to \{\{\delta V - V_i\}, \delta V - V\}$. which combines both inversion and a translation with $\delta V = 7$. That is the distribution for V will be the same as that for $7 - V$ and $\beta \to -\beta$. This is why the value of β for $\Sigma = 4D = 40$ matches that of $\Sigma = (7-4)D = 3D = 30$ just with a reversed sign.

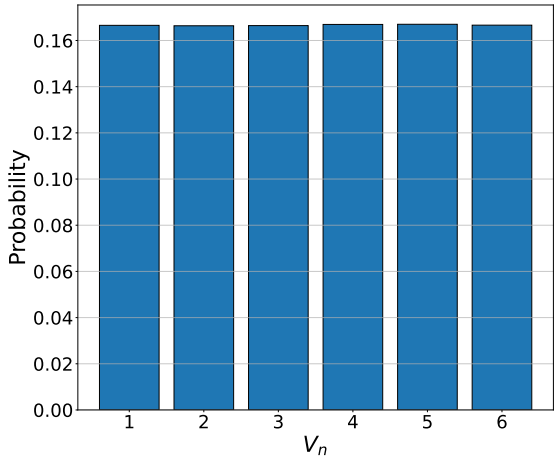


FIG. 2. The probability (relative frequency) of occurrence of each face value V_n over the $N_T = 100,000$ throws of $D = 10$ dice shown in Fig. 1. They are all $1/6$ as predicted for unconstrained systems. The equi-distribution is simply the consequence of the dice being fair.

Despite demonstrating a purely statistical understanding of all aspects of Fig. 1, a fundamental question still remains. Why does a purely statistical system know anything about entropy being maximized? This is tackled in the next subsection.

A. Why does the dice act like a system with maximized entropy?

Viewing the Σ-limited dice problem as a constrained system, one can now appreciate the Boltzmann-result of Fig. 1(right). It is, however, bizarre that this should happen. The Boltzmann solution is for equilibrium, that is, when entropy is maximized. There are no physical processes driving a 'dice dynamics'. Further, nothing is telling the system to maximize its entropy. The system of D dice is not bound by the second law of thermodynamics, so to speak. So, why does it act like so?

The resolution to this lies in the realization that the constrained system is 'derived' from the unconstrained system by means of sampling/filtering. If the unconstrained system has an equally likely probability for all states/configurations, then it is in a state of maximized entropy (equilibrium). Any sampling from it, inherits the fact that the entropy was already maximized. The dice being fair inherently implements the maximum entropy behavior, which is retained even when we constrain the system (sample from it). This is why the face-value distribution is the equilibrium

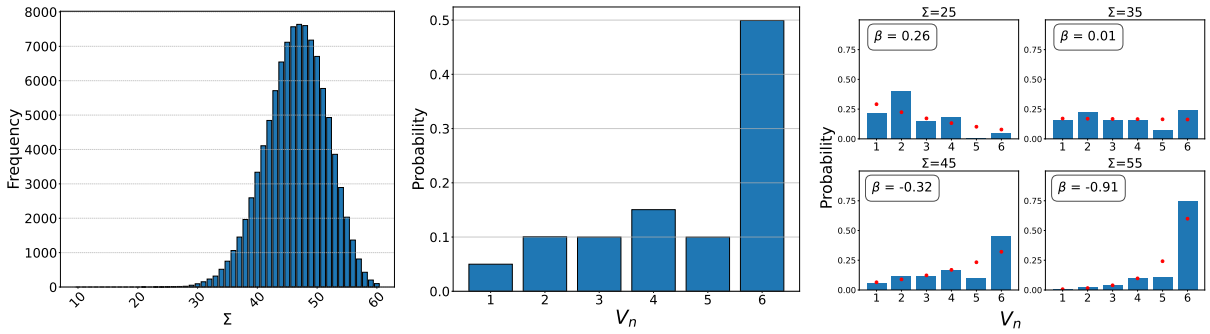


FIG. 3. Left: Histogram of all the possible sums Σ observed over $N_T = 100,000$ throws of $D = 10$ identical loaded dice. Middle: the distribution of the non-uniform appearance of face values of the dice. Right: Probability (relative frequency) of face values $V_n \in \{1,2,3,4,5,6\}$ for the subset of throws where $\Sigma = 25,35,45,55$. The red dots are the Boltzmann predictions.

state Boltzmann solution.

When an unconstrained system visits all its states equally, we say that it is 'Ergodic'. It does not matter how the system changes its state, but if the protocol for the change-of-state results in a uniform traversal of $\mathscr{P}_S$, it is grounds for modeling using the maximum entropy idea. Systems that are not ergodic[18], are not inherently in the maximized entropy state and a Boltzmann distribution is not guaranteed. This idea can also be demonstrated using the dice analogy. A non-ergodic system here would map to a loaded dice. If we repeat the steps of Fig. 1 we find the result shown in Fig. 3. The histogram (left panel) is not centered around the mean value of the faces, the face values themselves do not show up uniformly showing non-ergodic behavior (middle panel), and the measured frequencies for fixed Σ also do not line up with the Boltzmann prediction (right panel).

Note: On the ergodic requirement, Jaynes would have argued that for a seemingly non-ergodic distribution, it should be possible to find a parent ergodic system with suitable constraints that can reproduce the non-ergodic distribution. This possibility remains, but it still supports the idea that one needs a parent uniform distribution to sample from.

Thus, irrespective of the rule that evolves the state/configuration of the system (like natural dynamics of many bodies, or just performing re-runs of a random variable), it is the tendency to be ergodic that maximizes entropy. This is a pure statistical concept and can apply to systems beyond those that are governed by physical laws. To demonstrate this point, in part II of the article, we

will demonstrate thermodynamic principles like heat transfer, equilibration of two systems, and the second law of thermodynamics using the same system of dice.

V. CONCLUSION

The main purpose of this part of the article was to look at statistical mechanics in more general terms than what is usually portrayed in textbooks. To do so, it was necessary to distance ourselves from conventional terminology. Because of this more abstract view of statistical mechanics ideas, it is apparent that:

1. Entropy is just a measure of multiplicity of possibilities would map to the conventional idea of microstates of a system. These states are useful to imagine and construct a model for a system, but are often impractical for observational purposes.

2. The value-prescription is an observer dependent choice. It should be something that could be used to filter out the microstates. These value-based states would map to the conventional idea of the (practically observable) macrostates of a system, and the value would correspond to the energy. The filtering action is nothing but the assertion of conservation of energy (the first law of thermodynamics).

3. The notion of equilibrium state is simply the maximum entropy state which in turn is the most likely state due to high multiplicity.

4. The notion of ergodicity can be thought of as referring to the system's (and/or observation's) intrinsic tendency to equally sample the state-space when there are no constraining filters on the system.

5. The Boltzmann distribution maximizes the possibilities given a constraint on the mean value of an ergodic system, and is characterized by a parameter β.

We applied our abstraction to a system of fair dice and showed that:

- The fairness of the dice $\leftrightarrow$ ergodic behavior, which in turn implies an entropy maximized state. Any further sampling will not affect the underlying ergodicity and hence they always act like their entropy is maximized.

- If you constrain the sum of face values of the dice, this effectively samples the states consistent with the constraint and thus the distribution of the face value is the Boltzmann distribution which is known to maximize the entropy.

It is fitting to comment here that one hears several interpretations of entropy like it being a measure of disorder, uncertainty, and information capacity. We can unify all of these versions under the statement that entropy is a measure of allowed possibilities. It is the observer that may choose to classify the multitude of possibilities as intractable details (disorder), uncertainty, or multiple states to map information onto (information capacity).

A. Mapping the statistical ideas to information theory elements

To further demonstrate the utility of the above interpretation, we conclude by presenting a comparison of what we introduced here regarding the concept of entropy to that in information theory. Shannon entropy[19] in information theory is also defined like our $\tilde{S}$ in Eq. (1). The various parameters/aspects of the statistical entropy formulation and their interpretation in information theory are compared in Table I. It shows the analog of all the parameters or concepts we introduced: $N_T, \mathscr{P}_S, \{p_n\}, S$, etc., in the information theory sense. Some noteworthy distinctions are:

- In the dice problem, the concept of D was necessary to meaningfully generate a variety of macrostates, which would not be possible with just one die. Nevertheless, because all dice were identical, we managed to just work with the entropy associated with the states of a singe die. For a file, the data can be broken down into bytes. A byte already has 256 possibilities and hence it is sufficient to have $D = 1$.

- The base of the logarithm for information theory is 2. Observe that this is simply a multiplying factor of $1/\ln 2$ from the statistical entropy. We already saw that the overall multiplicative factor for entropy is often immaterial for its maximization. Nevertheless, in the statistical case, the natural log enters through the Sterling's approximation for the number of ways to realize an observed distribution. In the information theory context, the interest is in encoding the information using binary bits. Thus, if there are $\mathscr{W}$ configurations, then $\mathscr{W}$ units of information could be mapped to it. Since each bit can encode 2 units of information, to encode $\mathscr{W}$ units of information, one would need $\log_2 \mathscr{W}$ bits. This base of 2 is a choice that enables mapping the value to entropy to something physical: number of bits needed

parameter/concept	dice statistics	information theory
N_T	# re-runs (sample size)	# bytes (file size)
D	Number of identical constituents $= 10$	1
S	$-D\sum_n^{N_{\mathscr{P}_S}} p_n \ln p_n$	$-\sum_n^{N_{\mathscr{P}_S}} p_n \log_2 p_n$
$\mathscr{P}_S$	$\{1,2,3,4,5,6\}$	2^8 configurations of a byte
$N_{\mathscr{P}_S}$	6	$2^8 = 256$
$\{p_n\}$	frequency of states in the re-runs	frequency of byte-configuration in file
$\mathscr{W} \equiv$# possible ways to realize $\{p_n\}$	$e^{N_T S}$	$2^{N_T S}$
$\rightarrow$ high	many configurations	many information states
$\rightarrow$ max	equally sampled configurations	all configurations mapped to information
$\rightarrow$ interpretation	high likelihood of occurrence	high capacity for information
Constrained p_n	$p_n \propto e^{-\beta \times \text{value}}$	$p_n \propto e^{-\beta \times \text{distortion}}$
β	produces a specified mean value	produces a specified distortion

TABLE I. The various statistical aspects behind the idea of entropy and their interpretation in the statistical formulation vs the information theory formulation. They have the same core mathematical structure, but their interpretations differ due to the context of the respective problems.

to encode $\mathscr{W}$ units of information. In any case, the unifying notion is that $\mathscr{W}$ counts the multiplicity of different states.

- In information theory, S is sometimes presented as a measure of channel capacity and sometimes as the maximum limit of lossless compression of data. We can harmonize these interpretations easily with our perspective. Computing S for a given file gives the expected number of bits needed per byte of the file to encode the information it has. That is, the information $\mathscr{W} = 2^{N_T S}$ (where N_T is the file size). So, a channel with certain transmission entropy, can only transmit $\mathscr{W}$ units of information hence the term 'channel capacity'. Additionally, for the maximum lossless compression of $\mathscr{W} = 2^{N_T S}$ units of information, the new file size of size N_{comp} must have an entropy of $\text{Max}[-\sum p_n \ln p_n] = 8$. This must encode the same information $\mathscr{W}$. Thus, $\mathscr{W} = 2^{N_T S} = 2^{N_{\text{comp}} \times 8} \Rightarrow N_{\text{comp}} = N_T S/8$. Thus, S also directly gives the maximum size of the compressed file without losses.

- So far, no 'constraints' were placed on the entropy. A constrained $\{p_n\}$ in the statistical case corresponded to constraining the expected value of the system. In information theory[20], if the 'value' could refer to a measure of an error or distortion in the state of the byte, then the constraint on the value would amount to the extent of tolerated distortion in the data. Such a formulation is invoked to characterize lossy compression[21]. The value of S itself will indicate the number of bits needed to represent the compressed file.

Something that is not explicit in the table is the action of maximizing entropy. In the statistical sense, it is either an intrinsic property of the system, or is something one imposes to ensure drawing of the most unbiased inference (Jaynesian epistemology). In the information theory side, the notion of maximized entropy is not an intrinsic property of the system but is a design goal to pack the most information.

We thus see that the interpretation we provided also harmonizes the concepts of information theory with statistical physics, despite the former using entropy in markedly different manner than how it is used in physical systems.

REFERENCES

[1]W. M. Saslow, "A history of thermodynamics: The missing manual," Entropy **22**, 77 (2020).

[2]S. Carnot, *Réflexions sur la puissance motrice du feu et sur les machines propres à développer cette puissance* (Bachelier, Paris, 1824).

[3]J. P. Joule, "On the mechanical equivalent of heat," Philosophical Transactions of the Royal Society of London **140**, 61–82 (1850).

[4]R. Clausius, "Ueber die bewegende kraft der wärme und die gesetze, welche sich daraus für die wärmelehre selbst ableiten lassen," Annalen der Physik **155**, 368–397 (1850).

[5]W. Thomson, "On the Dynamical Theory of Heat, with Numerical Results Deduced from Mr Joule's Equivalent of a Thermal Unit, and M. Regnault's Observations on Steam," Transactions of the Royal Society of Edinburgh **20**, 261–288 (1851).

[6]O. Darrigol and J. Renn, "The emergence of statistical mechanics," in *The Oxford Handbook of the History of Physics*, edited by J. Z. Buchwald and R. Fox (Oxford University Press, 2013).

[7]E. T. Jaynes, "Information theory and statistical mechanics," Physical Review **106**, 620–630 (1957).

[8]E. T. Jaynes, "Information theory and statistical mechanics. ii," Physical Review **108**, 171–190 (1957).

[9]S. A. Frank, "The common patterns of nature," Journal of Evolutionary Biology **22**, 1563–1585 (2009).

[10]E. T. Jaynes, "Information theory and statistical mechanics," in *Statistical Physics: Brandeis Summer Institute Lectures in Theoretical Physics*, Vol. 3, edited by K. W. Ford (W. A. Benjamin, Inc., New York, 1963) pp. 181–218.

[11]E. T. Jaynes, "Information theory and statistical mechanics (1962 brandeis lectures)," in *E. T. Jaynes: Papers on Probability, Statistics and Statistical Physics*, edited by R. D. Rosenkrantz (D. Reidel Publishing Company, Dordrecht, Holland, 1983) pp. 39–76.

[12]M. J. Duff, L. B. Okun, and G. Veneziano, "Trialogue on the number of fundamental constants," Journal of High Energy Physics **2002**, 023 (2002).

[13]M. Duff, "How fundamental are fundamental constants?" Contemporary Physics **56**, 35–47 (2015), https://doi.org/10.1080/00107514.2014.980093.

[14]S. Maiti, "Harmonized units, a twist on natural units, and its implications," Research Square (2026).

[15]This can also be formulated in terms of a probability density, but let us retain the discrete form for this article.

[16]This may be derivable from the underlying states s_n of individual constituents, but is not of concern at the moment.

[17]L. D. Landau and E. M. Lifshitz, *Statistical Physics: Part 1*, 3rd ed., Vol. 5 (Pergamon Press, Oxford, UK, 1980).

[18]Whether a system is ergodic or not depends a lot τ_{Obs}. If it is too short, we simply may not have had enough wait-time to allow the system to visit all states.

[19]C. E. Shannon, "A mathematical theory of communication," The Bell System Technical Journal **27**, 379–423 (1948).

[20]Information theory, here uses 'mutual information' which results is a modified definition of the entropy functional.

[21]K. Rose, "A mapping approach to rate-distortion computation and analysis," IEEE Transactions on Information Theory **40**, 1939–1952 (1994).

[22]Y. Bengio, "Learning deep architectures for AI," Foundations and Trends® in Machine Learning **2**, 1–127 (2009).

[23] A. Hanna, https://github.com/YungHairStyle/Dice_simulation_Entropy_demo.